
Algorithm 1: Binary Atom Search Optimization (BASO) with True PRLU

Input: Population size N , Max iterations T , Parameters α, β , Dataset with F features.

Output: Optimal binary feature subset $\mathbf{X}_{gb}$.

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1 Initialize binary population  $\mathbf{X} \in \{0, 1\}^{N \times F}$  randomly;
2 Initialize velocities  $\mathbf{V} = \mathbf{0}^{N \times F}$  and set  $V_{max} = 6$ ;
3 Evaluate fitness for all atoms and find global best  $\mathbf{X}_{gb}$  and  $Fit_{gb}$ ;
4 for  $t = 1$  to  $T$  do
5   Calculate mass  $M_i(t)$  for all atoms using Eq. (2);
6   Determine  $K_{best}(t)$  using Eq. (3);
7   Update time-varying parameters  $\eta(t)$  and  $\lambda(t)$  using Eq. (4) and (5);
8   for each atom  $i = 1$  to  $N$  do
9     Calculate length scale  $\sigma_i$  relative to centroid of  $K_{best}$  atoms (Eq.
10    6);
11     $\mathbf{A}_i \leftarrow \mathbf{0}$ ; // Initialize acceleration
12    for each atom  $j \in K_{best}$  do
13      Calculate Euclidean distance  $r_{ij} = \|\mathbf{X}_i - \mathbf{X}_j\|$ ;
14      Calculate normalized distance
15       $h_{ij} = \text{clamp}\left(\frac{r_{ij}}{\sigma_i + \epsilon}, h_{min}, h_{max}\right)$ ;
16      Compute Lennard-Jones force magnitude
17       $F_{mag} = -\eta(t) \cdot (2h_{ij}^{13} - h_{ij}^7)$ ;
18      Generate random weight  $r_j \sim U(0, 1)$ ;
19      // Calculate total acceleration using Eq. 9
20      Calculate total acceleration  $\mathbf{A}_i$  using Eq. (9);
21      for each dimension  $d = 1$  to  $F$  do
22        // Update velocity (Eq. 10)
23         $V_i^d \leftarrow \text{clip}(rand(0, 1) \cdot V_i^d + A_i^d, -V_{max}, V_{max})$ ;
24        // Map to binary position using True PRLU
25        Calculate probability  $S = f_{\text{TruePRLU}}(V_i^d)$ ;
26         $X_i^d \leftarrow 1$  if  $rand(0, 1) < S$  else 0;
27      // Re-evaluate fitness and update global best
28      Evaluate fitness for all atoms;
29      if new best fitness found then
30        Update  $\mathbf{X}_{gb}$  and  $Fit_{gb}$ ;
31 return  $\mathbf{X}_{gb}$ ;
  
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